

Max. Marks: 100

4. Speed of light $c = 3 \times 10^8 \text{ m/s}$, $K = 1.38 \times 10^{-23} \text{ J/K}$, $h = 6.625 \times 10^{-34} \text{ JS}$, $g = 9.8 \text{ m/s}^2$, $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$

Module – 1			M	L	C
Q.1	a.	State and explain Heisenberg's uncertainty principle and show that there is no existence of electrons in the nucleus of an atom.	9	L2	CO1
	b.	What is a wave function, probability density and normalization of wave function?	7	L2	CO1
	c.	Find the lowest energy of an electron confined to move in a one dimensional potential box of length 1A in electron volts.	4	L3	CO1
OR					
Q.2	a.	Setup time Independent Schrodinger's wave equation for a particle in one dimension.	7	L2	CO1
	b.	Discuss the wave functions, probability densities and energy for a particle in a box by considering the ground state and first two excited states.	9	L2	CO1
	c.	Calculate the de-Broglie wavelength of an electron when it is accelerated to a potential of 5000 V.	4	L3	CO1
Module – 2					
Q.3	a.	Mention any three assumptions of quantum free electron theory. Discuss the variation of Fermi factor with temperature and energy.	9	L2	CO1
	b.	Explain the construction and working of MAGLEV vehicle.	6	L2	CO1
	c.	An elemental solid dielectric material has polarizability of $7 \times 10^{-40} \text{ Fm}^2$. Assuming the internal field to be Lorentz field, calculate the dielectric constant for the material if the material has $3 \times 10^{28} \text{ atoms/m}^3$.	5	L3	CO1
OR					
Q.4	a.	What is super conductivity? Describe Type-I and Type-II superconductors.	7	L2	CO1
	b.	What is dielectric polarization? Explain various types of polarization mechanism.	8	L2	CO1
	c.	Calculate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 200 K and 400 K in a material.	5	L3	CO1
Module – 3					
Q.5	a.	Obtain an expression for energy density of radiation under thermal equilibrium conditions in terms of Einstein's coefficients.	8	L2	CO2
	b.	What is attenuation? Explain different types of attenuation in optical fibers.	8	L2	CO2
	c.	The average output power of laser source emitting a laser beam of wave length 6328 A is 5 mW. Find the number of photons emitted per second by the laser source.	4	L3	CO2
OR					

Q.6	a.	What is numerical aperture? Obtain an expression for numerical aperture in terms of refractive indices of core and cladding of an optical fiber.	9	L2	CO2
	b.	Describe the working of a laser printer.	6	L2	CO2
	c.	The attenuation of light in an optical fiber is estimated at 2.2 dB/km. What fractional initial intensity remains after 2 km and after 6 km.	5	L3	CO2

Module – 4

Q.7	a.	State and prove Gauss Divergence theorem.	7	L2	CO3
	b.	Explain Faraday's laws of electromagnetic induction and amperes law. Express the same in point form.	8	L2	CO3
	c.	Determine the constant c such that the vector $\vec{A} = (x + ay)\hat{a}_x + (y + bz)\hat{a}_y + (x + cz)\hat{a}_z$ is solenoidal.	5	L3	CO3

OR

Q.8	a.	Derive wave equation in terms of electric field using Maxwell's equations for free space.	8	L2	CO3
	b.	Discuss continuity equation. Derive the expression for displacement current.	8	L2	CO3
	c.	Calculate the curl of $\vec{A}$ given by $\vec{A} = (1 + yz^2)\hat{a}_x + xy^2\hat{a}_y + x^2y\hat{a}_z$.	4	L3	CO3

Module – 5

Q.9	a.	Derive an expression for electrical conductivity in extrinsic and intrinsic semiconductors.	8	L2	CO4
	b.	Describe the construction and working of semiconductor laser with energy level diagram.	8	L2	CO4
	c.	The Hall coefficient of a specimen of a doped silicon is found to be $3.66 \times 10^{-4} \text{ m}^3/\text{c}$. The resistivity of the specimen is $9.93 \times 10^{-3} \text{ ohm-m}$. Find the mobility and charge carrier density assuming single carrier conduction.	4	L3	CO4

OR

Q.10	a.	Explain Fermi level in an intrinsic semiconductor and derive the relation between Fermi energy and energy gap for an intrinsic semiconductor.	9	L2	CO4
	b.	Explain construction and working of photo diode.	7	L2	CO5
	c.	The resistivity of intrinsic germanium at 27°C is 0.47 ohm-meter . If the electron and hole mobilities are $0.38 \text{ m}^2/\text{VS}$ and $0.18 \text{ m}^2/\text{VS}$ respectively. Calculate the intrinsic carrier density.	4	L3	CO4

MAKE-UP EXAM



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BPHYE102/202

First/Second Semester B.E./B.Tech. Degree Examination, Nov./Dec. 2023 Applied Physics for EEE Stream

Time: 3 hrs.

Max. Marks: 100

- Note:** 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	State Heisenberg's uncertainty principle. Show that an electron does not exist inside the nucleus on the basis of Heisenberg's uncertainty principle.	7	L2	CO1
	b.	Starting from Schrodinger's time independent wave equation, derive the expression for energy eigen value and eigen function for an electron in one dimensional potential well of infinite height.	8	L2	CO1
	c.	A particle of mass $0.5 \text{ MeV}/c^2$ has kinetic energy 100eV . Find its de Broglie wavelength, where C is the velocity of light.	5	L3	CO1
OR					
Q.2	a.	Set up time independent Schrödinger wave equation for a free particle in one dimension.	7	L2	CO1
	b.	Explain probability density and normalization. Give the physical significance of a wave function.	8	L2	CO1
	c.	In a measurement that involved an inherent uncertainty of 0.003% , the speed of an electron was found to be 800m/s . Calculate the corresponding uncertainty involved in determining its position.	5	L3	CO1
Module – 2					
Q.3	a.	Explain polarization of dielectric. Discuss the mechanism of different types of polarization.	9	L2	CO1
	b.	Describe the type I and type II superconductors.	6	L2	CO1
	c.	Find the temperature at which there is 1% probability that a state with an energy 0.5eV above Fermi energy is occupied.	5	L3	CO1
OR					
Q.4	a.	Mention any two assumption of quantum fee theory? Discuss the probability of occupation of various energy states by electron on temperature on the basis of Fermi factor.	9	L2	CO1
	b.	Explain the construction and working of Maglev vehicles.	6	L2	CO1

	c.	The dielectric constant of Helium gas at NTP is 1.0000684. Calculate the electronic polarizability of the atoms if Helium gas contains 2.7×10^{25} atoms/m ³ and kept in an electric field of 3×10^4 V/m.	5	L3	CO1
CMRIT LIBRARY BANGALORE - 560 037 Module – 3					
Q.5	a.	Obtain an expression for energy density of radiation under thermal equilibrium condition in terms of Einstein's co-efficient.	8	L2	CO2
	b.	What is attenuation? Discuss different types of attenuation in optical fiber.	7	L2	CO2
	c.	The ratio of population of upper to lower energy levels is 1.059×10^{-30} . Find the wave length of light emitted by spontaneous emission at 300K.	5	L3	CO2
OR					
Q.6	a.	Define Numerical aperture. Obtain an expression for numerical aperture in an optical fiber.	8	L2	CO2
	b.	Describe the principle, construction and working of carbon dioxide laser with energy level diagram.	7	L2	CO2
	c.	In a step index optical fiber with a core diameter of 60μm and core and cladding refractive indices as 1.5 and 1.48 respectively. When the wavelength of 850nm is propagating through it. Calculate the numerical aperture, fractional index change, V-parameter and number of modes in the fiber.	5	L3	CO2
Module – 4					
Q.7	a.	Discuss about continuity equation. Derive the expression for displacement current.	8	L2	CO3
	b.	Explain Faraday's law of Electromagnetic induction, Amperes law and express the same in point form.	7	L2	CO3
	c.	Determine the constant 'C' such that, the vector $\vec{A} = (x + ay)\hat{a}_x + (y + bz)\hat{a}_y + (x + cz)\hat{a}_z$ is solenoidal.	5	L3	CO3
OR					
Q.8	a.	Derive wave equation for electromagnetic waves in vacuum in terms of electric field using Maxwell's equation.	8	L2	CO3
	b.	What are vector operator ∇ and explain the concept of divergence, gradient and curl.	7	L2	CO3
	c.	Calculate the curl of $\vec{A}$, given $\vec{A} = (1 + yz^2)\hat{a}_x + xy^2\hat{a}_y + x^2y\hat{a}_z$	5	L3	CO3

Module – 5					
Q.9	a.	Discuss the law of mass action. Establish relation between Fermi energy and energy gap for an intrinsic semiconductor.	8	L2	CO4
	b.	Define Hall voltage and Hall field? Obtain expression for Hall voltage in term of Hall co-efficient.	7	L2	CO4
	c.	The resistivity of intrinsic germanium at 27°C is equal to 0.47 ohm-meter. Assuming electron and hole mobilities as $0.38\text{m}^2\text{V}^{-1}\text{s}^{-1}$ and $0.18\text{m}^2\text{V}^{-1}\text{S}^{-1}$ respectively. Calculate the intrinsic carrier density.	5	L3	CO4
OR					
Q.10	a.	Derive an expression for electrical conductivity in extrinsic and intrinsic semiconductor.	8	L2	CO4
	b.	Explain the construction and working of photo transistor and also mention any two applications.	7	L2	CO4
	c.	In diffraction gravity experiment the laser light undergoes second order diffraction for diffraction angle 1.48° . The grating constant $d = 5.05 \times 10^{-5}\text{m}$ and the distance between the grating and screen is 0.60m. Calculate the wavelength of laser source.	5	L3	CO5

CBCS SCHEME

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BPHYE102/202

First/Second Semester B.E./B.Tech.Degree Examination, June/July 2023 Applied Physics for EEE Stream

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	What is Wave function? Setup Time Independent Schrodinger wave equation for a Quantum free particle in one dimension and mention the same in three dimensions.	9	L2	CO1	
	b.	Explain de-Broglie hypothesis. Arrive at the expression for de-Broglie wavelength.	6	L2	CO1	
	c.	Calculate the energy in first two energy states for an electron in one dimensional infinite potential well of width 1 nm.	5	L3	CO1	
OR						
Q.2	a.	Starting from Schrodinger Time Independent wave equation, obtain the expressions for Energy Eigen values and normalized Eigen wave function for a particle in one dimensional infinite potential well.	9	L2	CO1	
	b.	State and explain Heisenberg's uncertainty principle. Using the same, show the non-existence of electron inside the nucleus.	6	L2	CO1	
	c.	An electron and a photon each have same de-Broglie wavelength of 10 Å. Find the kinetic energy of electron and energy of photon.	5	L3	CO1	
Module – 2						
Q.3	a.	Define superconductivity and critical temperature. Explain briefly SQUID and mention its any three applications.	10	L2	CO2	
	b.	Mention the expression for internal field for one dimensional array of atoms in case of solid dielectrics. Derive Clausius Mossotti equation.	6	L2	CO2	
	c.	Evaluate the probability of an electron occupying an energy level 0.02 eV above the Fermi level at 400 K.	4	L3	CO2	
OR						
Q.4	a.	Explain Electronic, Ionic and Orientation polarizations of dielectrics. Outline briefly the solid and liquid dielectrics with one example each.	10	L2	CO2	
	b.	Give any four assumptions of quantum free electron theory. Mention the expression for electrical conductivity in metals and explain the symbols.	6	L2	CO2	
	c.	Lead in the superconducting state has critical temperature of 6.2 K at zero magnetic field and a critical field of 0.064 MA ⁻¹ at 0 K. Determine the critical field at 4 K.	4	L3	CO2	
Module – 3						
Q.5	a.	Mention any four characteristics of LASER and explain three types of interaction of radiation with matter.	10	L2	CO1	

BPHYE102/202					
	b.	With a neat labelled block diagram, explain the application of optical fibers in point to point communication.	6	L2	CO1
	c.	A medium in thermal equilibrium at 300 K has two energy levels with wavelength separation $1\text{ }\mu\text{m}$. Find the population densities ratio of upper and lower energy levels.	4	L3	CO1
OR					
Q.6	a.	Derive an expression for numerical aperture of an optical fiber. Discuss step-index single mode and multi mode optical fibers.	10	L2	CO1
	b.	Discuss the application of laser in laser range finder in Defence.	6	L2	CO1
	c.	The sum of refractive indices of core and cladding in a step index optical fiber is 3.12 and their difference is 0.08. The diameter of core is $50\text{ }\mu\text{m}$. Find the V-number and number of modes that the fiber can support for propagation at the wavelength 594 nm.	4	L3	CO1
Module – 4					
Q.7	a.	Write Maxwell's equations in vacuum. Drive equation of electromagnetic waves using Maxwell's equation in free space.	8	L2	CO3
	b.	State and derive Gauss divergence theorem. Mention Stoke's theorem in mathematical form.	7	L2	CO3
	c.	Find the resonance frequency of an LCR series circuit with inductance = 0.5 henry, capacitance = 0.45 microfarad and resistance = 400 ohm.	5	L3	CO5
OR					
Q.8	a.	State and explain : Gauss law in magnetism, Ampere's circuital law and Biot-Savart's law.	8	L2	CO3
	b.	Discuss continuity equation. Derive an expression for displacement current.	7	L2	CO3
	c.	Find the wavelength of the semiconductor laser in the diffraction grating experiment when the angle of diffraction is 1.5 degree for the second order maximum. Given grating constant = 4.7×10^{-5} per metre.	5	L3	CO5
Module – 5					
Q.9	a.	State and explain law of mass action for semiconductors. Derive the expression for Fermi energy in terms of energy gap of intrinsic semiconductor.	10	L2	CO4
	b.	Describe the construction and working of semiconductor laser diode.	6	L2	CO4
	c.	The concentration of electrons at room temperature in an extrinsic semiconductor used in Hall experiment is $1.7 \times 10^{29}/\text{m}^3$. If a current of 100 A flows through the semiconducting strip of thickness 1 mm and the strength of magnetic field perpendicular to the strip is 3.6 wb/m^2 , estimate the Hall coefficient and Hall voltage.	4	L3	CO4
OR					
Q.10	a.	What is Hall effect? Derive an expression for Hall voltage in terms of Hall coefficient. Mention any three applications of Hall effect.	10	L2	CO4
	b.	Explain Four probe method to determine resistivity of semiconductor.	6	L2	CO4
	c.	Mobilities of electrons and holes in a sample of intrinsic Germanium at 300 K are $0.36\text{ m}^2\text{V}^{-1}\text{s}^{-1}$ and $0.14\text{ m}^2\text{V}^{-1}\text{s}^{-1}$ respectively. If the resistivity of Ge is $2.5\text{ }\Omega\text{m}$, compute the intrinsic carrier density.	4	L3	CO4

CBCS SCHEME

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BPHYE102

First Semester B.E./B.Tech. Degree Examination, Jan./Feb. 2023

Applied Physics for EEE Stream

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. *M* : Marks , *L*: Bloom's level , *C*: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define Wave function, then setup time – independent one dimensional Schrodinger's wave equation.	8	L2	CO1
	b.	State Heisenberg's uncertainty principle and show that a free electron cannot exist within the nucleus of an atom.	7	L2	CO1
	c.	Calculate the de Broglie wavelength associated with an electron with a kinetic energy of 1.5ev.	5	L2	CO1
OR					
Q.2	a.	Derive Eigen function equation for a particle in one – dimensional potential wall of infinite height.	8	L2	CO1
	b.	Explain de Broglie Hypothesis. Derive the expression for de Broglie wavelength and mention different forms of de Broglie wavelength.	7	L2	CO1
	c.	In a measurement of position and momentum that involved an uncertainty of 0.003% , the speed of an electron was found to be 800 m/s. Calculate the corresponding uncertainty that arises in determining its position.	5	L3	CO1
Module – 2					
Q.3	a.	What is Fermi factor and explain the variation of Fermi factor with respect to temperature and energy.	8	L2	CO2
	b.	Define Internal field and derive Clausius - Mossotti relation.	7	L2	CO2
	c.	Solve the polarization produced in crystal by an electric field of strength 500 vm^{-1} , If it has a dielectric constant of 6.	5	L3	CO2
OR					
Q.4	a.	Classify and explain Type – I and Type – II superconductors.	8	L2	CO2
	b.	Define Dielectric Polarization and explain different types of Electrical polarization mechanisms.	7	L2	CO2
	c.	Calculate the probability of an electron occupying and energy level 0.02ev above the Fermi level at 200K and 400K.	5	L3	CO2
Module – 3					
Q.5	a.	Define Spontaneous Emission and Stimulates Emission. Derive the expression for energy density of radiation at equilibrium in terms of Einstein's coefficients.	8	L2	CO1
	b.	Describe different types of optical fibres , with neat diagram.	7	L2	CO1
	c.	In a optical fiber experiment, the distance between the fiber and screen is 45mm and radius of the spot is 6mm. Calculate the angle of acceptance and numerical aperature of an optical fiber.	5	L3	CO5
1 of 2					

OR

Q.6	a.	Define Numerical aperture and derive the expression for numerical aperture of an optical fiber and mention the condition for ray propagation in optical fiber.	8	L2	CO1
	b.	With neat diagrams, explain the construction and working of CO ₂ Laser.	7	L2	CO1
	c.	The average output power at Laser source emitting a Laser Beam of wavelength 632.8nm is 5mw. Find the number of photons emitted per second by the Laser source.	5	L3	CO1

Module – 4

Q.7	a.	State and explain Gauss Divergence theorem and mention the Stoke's theorem.	8	L2	CO3
	b.	Derive Wave equation in terms of electric field using Maxwell's equation for free space.	7	L2	CO3
	c.	A circular coil of wire consisting of 100 turns. Each of radius 8cm carries a current of 0.4A. What is the magnitude of the magnetic field at a point 20cm from the wire?	5	L3	CO5

OR

Q.8	a.	What is Displacement current? Derive the expression for displacement current.	8	L2	CO3
	b.	State Biot – Savart's law and list four Maxwell's equations in differential form.	7	L2	CO3
	c.	Find constant C, such that $\vec{A} = (x + ay) \hat{a}_x + (y + bz) \hat{a}_y + (x + cz) \hat{a}_z$ is Solenoid.	5	L3	CO3

Module – 5

Q.9	a.	What is Hall Effect? Obtain the expression for Hall coefficient and express Hall Voltage in terms of Hall coefficient.	8	L2	CO4
	b.	Explain Construction and working of Photodiode and mention its applications.	7	L2	CO4
	c.	In a Plank's constant experiment, the knee voltage of a given Red LED emitting light of wavelength 650nm is 1.908V. Calculate the Plank's constant.	5	L3	CO5

OR

Q.10	a.	Explain Construction and working of an Semiconductor Laser and mention its applications.	8	L2	CO4
	b.	Mention the expression for Hole and Electron concentration in a intrinsic semiconductor and derive the expression for electrical conductivity of a semiconductor.	7	L2	CO4
	c.	The Hall coefficient of a material is $-3.68 \times 10^{-5} \text{ m}^3/\text{C}$. What is the type of charge carriers? Also calculate the carrier concentration.	5	L3	CO4

Model Question Paper-I with effect from 2022-23 (CBCS Scheme)

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First/Second Semester B.E. Degree Examination

Subject Title APPLIED PHYSICS FOR EEE STREAM

TIME: 03 Hours

Max. Marks: 100

- Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
 02. Draw neat sketches where ever necessary
 03. **Constants** : Speed of Light ' c ' = $3 \times 10^8 \text{ ms}^{-1}$, Boltzmann Constant ' k ' = $1.38 \times 10^{-23} \text{ JK}^{-1}$,
 Planck's Constant ' h ' = $6.625 \times 10^{-34} \text{ Js}$, Acceleration due to gravity ' g ' = 9.8 ms^{-2} ,
 Permittivity of free space ' ϵ_0 ' = $8.854 \times 10^{-12} \text{ F m}^{-1}$

Module -1			*Bloom's Taxonomy Level	Marks
Q.01	a	State and explain Heisenberg's uncertainty principle. Show that an electron does not exist inside the nucleus on the basis of Heisenberg's Uncertainty Principle.	L2	09
	b	What is wave function? Give its physical significance and properties	L2	06
	c	An electron is bound in a 1-dimensional potential well of width $1\text{\AA}$ & of infinite height. Find its energy values in eV in the ground state & also in the first two excited states.	L3	05
OR				
Q.02	a	Setup time independent Schrodinger wave equation for free particle in one dimension	L2	07
	b	Discuss the wave functions, probability densities and energy level for a particle in a box by considering the ground and the first two excited State	L2	09
	c	Calculate the deBroglie wavelength associated with an electron having a kinetic energy of 100 eV	L3	04
Module-2				
Q. 03	a	Explain the dependence of resistance on temperature of a superconductor? Describe Type I and type II superconductors	L2	09
	b	Derive Clausius Mossotti equation.	L2	06
	c	Show that the sum of the probability of Occupancy of an energy state at ΔE below Fermi level and that at ΔE above Fermi level is unity.	L3	05
OR				
Q.04	a	Mention any three assumptions of quantum free electron theory? Discuss the dependence of Fermi factor on temperature and Consequent effect on probability of occupation of energy levels	L2	09
	b	Explain the construction and working of MAGLEV vehicle.	L2	06
	c	The dielectric constant of helium gas at NTP is 1.0000684. Calculate the electronic polarizability of the atoms if helium gas contains 2.7×10^{25} atoms/ m^3 .	L3	05
Module-3				
Q. 05	a	Describe the Principle, Construction and Working of Carbon Dioxide Laser with energy level diagram.	L2	08
	b	What is numerical aperture. Obtain an expression for numerical aperture in an optical fiber.	L2	08
	c	A medium in thermal equilibrium at a temperature 300 K has two energy levels with wavelength separation of $1 \mu\text{m}$. Find the ratio of population densities of the upper & lower levels.	L3	04

OR				
Q. 06	a	Obtain an expression for energy density of radiation under thermal equilibrium condition in terms of Einstein's coefficients.	L2	08
	b	What is attenuation? Discuss different types of attenuation in optical fibers.	L2	07
	c	In a step index optical fiber with core diameter of 60 μm & core and cladding refractive indices as 1.50 & 1.48 respectively, when the wavelength of 850 nm is propagating through it. Calculate the numerical aperture, fractional index change, V parameter and number of modes in the fiber.	L3	05
Module-4				
Q. 07	a	State and prove Gauss Divergence theorem	L2	07
	b	Derive wave equation in terms of electric field using Maxwell's equation for free space	L2	08
	c	Prove that $3y^4 z^2 \hat{x} + 4x^3 z^2 \hat{y} + 3x^2 y^2 \hat{z}$ is Solenoidal.	L3	05
OR				
Q. 08	a	What is displacement current? Derive the expression for displacement current	L2	07
	b	Explain Faraday's Laws of Electromagnetic induction, Amperes Law and express the same in point form.	L2	08
	c	Elucidate the Transverse nature of EM Waves through Linear Polarization.	L3	05
Module-5				
Q. 09	a	Derive an expression for Electrical conductivity in extrinsic and intrinsic semiconductors.	L2	08
	b	Explain how the resistivity of a semiconductor is determined using four probe method? Mention any two applications of four probe method.	L2	07
	c	An n- type Germanium sample as a Donor density of $10^{21}/\text{m}^3$. it is arranged in a Hall experiment having magnetic field of 0.5 T and the current density is 500 A/m^2 . Find the Hall voltage if the sample is 3 mm wide.	L3	05
OR				
Q. 10	a	What is Hall effect? obtain an expression for the Hall coefficient	L2	08
	b	Explain the construction and working of photodiode? Discuss the power responsivity in a photodiode.	L2	08
	c	The resistivity of intrinsic Germanium at 27°C is equal to 0.47 ohm-metre . Assuming electron and hole mobilities as $0.38 \text{ m}^2\text{v}^{-1} \text{ s}^{-1}$ and $0.18 \text{ m}^2\text{v}^{-1} \text{ s}^{-1}$ respectively. Calculate the intrinsic carrier density.	L3	04

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

Table showing the Bloom's Taxonomy Level, Course Outcome and Program Outcome				
Question		Bloom's Taxonomy Level attached	Course Outcome	Program Outcome
Q.1	(a)	L2	1	1,2
	(b)	L2	1	1,2,12
	(c)	L3	1	1,2
Q.2	(a)	L2	1	1,2,12
	(b)	L2	1	1,2,12
	(c)	L3	1	1,2
Q.3	(a)	L2	1	1,2
	(b)	L2	1	1,2
	(c)	L3	1	1,2
Q.4	(a)	L2	1	1,2
	(b)	L2	1	1,2,12
	(c)	L3	1	1,2
Q.5	(a)	L2	2	1,2,12
	(b)	L2	2	1,2,12
	(c)	L3	2	1,2
Q.6	(a)	L2	2	1,2
	(b)	L2	2	1,2,12
	(c)	L3	2	1,2
Q.7	(a)	L2	3	1,2,12
	(b)	L2	3	1,2,12
	(c)	L3	3	1,2
Q.8	(a)	L2	3	1,2,12
	(b)	L2	3	1,2,12
	(c)	L3	3	1,2
Q.9	(a)	L3	4	1,2,12
	(b)	L2	4	1,2,12
	(c)	L3	4	1,2
Q.10	(a)	L2	4	1,2,12
	(b)	L2	4	1,2,12
	(c)	L3	4	1,2

Model Question Paper-II with effect from 2022-23 (CBCS Scheme)

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First/Second Semester B.E. Degree Examination

Subject Title APPLIED PHYSICS FOR EEE STREAM

TIME: 03 Hours

Max. Marks: 100

- Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
 02. Draw neat sketches where ever necessary
 03. **Constants** : Speed of Light ' c ' = $3 \times 10^8 \text{ ms}^{-1}$, Boltzmann Constant ' k ' = $1.38 \times 10^{-23} \text{ JK}^{-1}$,
 Planck's Constant ' h ' = $6.625 \times 10^{-34} \text{ Js}$, Acceleration due to gravity ' g ' = 9.8 ms^{-2} ,
 Permittivity of free space ' ϵ_0 ' = $8.854 \times 10^{-12} \text{ F m}^{-1}$

Module -1			*Bloom's Taxonomy Level	Marks
Q.01	a	Obtain the expression for Eigen value and Eigen function for particle in a box.	L2	10
	b	What is wave function, Probability density and normalisation of wave function	L2	06
	c	An electron has a speed of $4.8 \times 10^5 \text{ m/s}$ accurate to 0.012%. With what accuracy can be located the position of electron.	L3	04
OR				
Q.02	a	Derive expression for deBroglie wavelength. Mention different forms for deBroglie wavelength.	L2	08
	b	State and explain Heisenberg Uncertainty principle and Principle of Complementarity.	L2	07
	c	A particle of mass $0.65 \text{ MeV}/c^2$ has kinetic energy 120 eV . Find its deBroglie wavelength, where c is the velocity of light.	L3	05
Module-2				
Q. 03	a	What is superconductivity? Outline, qualitatively the BCS theory of superconductivity.	L2	08
	b	What is polarisation? Describe the different polarisation mechanism.	L2	08
	c	If a NaCl crystal is subjected to an electric field of 1000 V/m and the resulting polarization is $4.3 \times 10^{-8} \text{ C/m}^2$. Calculate the dielectric constant of NaCl.	L3	04
OR				
Q.04	a	Discuss the solid, liquid and gaseous dielectrics with one example each. Explain the role of dielectrics as a electrical insulator.	L2	08
	b	What are SQUIDS? Explain briefly the working of SQUID.	L2	08
	c	At 6 K critical field is $5 \times 10^3 \text{ A/m}$. Calculate the transition temperature when critical magnetic field is $2 \times 10^4 \text{ A/m}$ at 0 K .	L3	04
Module-3				
Q. 05	a	Explain the conditions and requisites for Laser action.	L2	07
	b	Define numerical aperture and fraction index change. Explain different types of optical fibers.	L2	08
	c	Find the core radius necessary for single mode operation at 850 nm with core and cladding refractive indices 1.49 and 1.48 respectively, also calculate the number of the modes present in the fiber.	L3	05

OR				
Q. 06	a	Discuss point to point optical fiber communication system. Mention two advantages and disadvantages of optical fiber system.	L2	09
	b	Discuss the working of laser printer	L2	07
	c	He-Ne laser is emitting a laser beam with an average power of 4.5mW. Find the number of photons emitted per second by the laser. The wavelength of the emitted radiation is 632.8 nm	L3	04
Module-4				
Q. 07	a	Derive wave equation for electromagnetic waves in vacuum in terms of electric field using Maxwell's equation	L2	09
	b	Explain the terms gradient of a scalar, divergence and curl of a vector. Derive Gauss divergence theorem	L2	07
	c	Find the divergence of the vector field $\vec{A}$ given by $\vec{A} = 6x^2\vec{a}_x + 3xy^2\vec{a}_y + xyz^3\vec{a}_z$	L3	04
OR				
Q. 08	a	Discuss continuity equation? Derive the expression for displacement current.	L2	09
	b	Explain the transverse nature of electromagnetic waves.	L2	07
	c	Calculate the curl of $\vec{A}$ given by $\vec{A} = (1 + yz^2)\vec{a}_x + xy^2\vec{a}_y + x^2y\vec{a}_z$	L3	04
Module-5				
Q. 09	a	Establish relation between Fermi energy and energy gap for an intrinsic semiconductor. Discuss the law of mass action	L2	09
	b	What is Hall voltage & Hall field? Obtain expression for Hall voltage in terms of Hall Coefficient	L2	06
	c	An n- type Germanium sample as a Donor density of $10^{21}/\text{m}^3$. it is arranged in a Hall experiment having magnetic field of 0.5 T and the current density is 500 A/m ² . Find the Hall voltage if the sample is 3 mm wide.	L3	05
OR				
Q. 10	a	Describe the construction and working of semiconductor laser with energy level diagram.	L2	08
	b	Explain the construction and working of photo transistor and also mention any two applications.	L2	08
	c	The following data are given for intrinsic germanium at 300 K, $n_i = 2.4 \times 10^{19}/\text{m}^3$, $\mu_e = 0.39 \text{ m}^2\text{v}^{-1} \text{ s}^{-1}$, $\mu_h = 0.19 \text{ m}^2\text{v}^{-1} \text{ s}^{-1}$. Calculate the resistivity of the sample.	L3	04

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

Table showing the Bloom's Taxonomy Level, Course Outcome and Program Outcome				
Question		Bloom's Taxonomy Level attached	Course Outcome	Program Outcome
Q.1	(a)	L2	1	1,2,12
	(b)	L2	1	1,2,12
	(c)	L3	1	1,2
Q.2	(a)	L2	1	1,2
	(b)	L2	1	1,2,12
	(c)	L3	1	1,2
Q.3	(a)	L2	1	1,2,12
	(b)	L2	1	1,2,12
	(c)	L3	1	1,2
Q.4	(a)	L2	1	1,2
	(b)	L2	1	1,2
	(c)	L3	1	1,2
Q.5	(a)	L2	2	1,2,12
	(b)	L2	2	1,2,12
	(c)	L3	2	1,2
Q.6	(a)	L2	2	1,2,12
	(b)	L2	2	1,2,12
	(c)	L3	2	1,2
Q.7	(a)	L2	3	1,2,12
	(b)	L2	3	1,2,12
	(c)	L3	3	1,2
Q.8	(a)	L2	3	1,2
	(b)	L2	3	1,2,12
	(c)	L3	3	1,2
Q.9	(a)	L3	4	1,2,12
	(b)	L2	4	1,2,12
	(c)	L3	4	1,2
Q.10	(a)	L2	4	1,2,12
	(b)	L2	4	1,2,12
	(c)	L3	4	1,2